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Presentation on Topic: AI Powered Cyber Threat Detection, Classification & Alert System

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OUTLINE

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INTRODUCTION

The project is a cybersecurity solution designed to tackle the growing challenges of real-time threat detection, analysis, and response. Traditional tools like SIEMs and dashboards can visualize data but often fall short in providing autonomous intelligence and proactive defence. To bridge this gap, it integrates real-time data ingestion using Kafka, AI-driven threat summarization, and automated responses powered by agentic AI. By unifying monitoring, intelligent analysis, and rapid incident handling in a scalable pipeline, the system aims to enhance detection accuracy, reduce response latency, and strengthen the overall security posture of enterprises.

MOTIVATION

- **Evolving threat landscape:** Advanced cyberattacks outpacing traditional defence's
- **Reactive tools:** Current systems focus on alerts, not proactive action
- **Human dependency:** Heavy reliance on analysts slows down response
- **Intelligence gap:** Fragmented data makes it hard to extract actionable insights
- **AI potential:** Leverage machine learning for adaptive, autonomous defence
- **Need for speed:** Real-time ingestion and processing to stay ahead of attackers
- **Holistic security:** Unifying detection, analysis, and automated response

LITERATURE REVIEW

Journal Name	Year	Authors	Title of the paper	Algorithm used	Results & Conclusion	Research Gaps	Limitations/Challenges	Links
Internet of Things and Cyber Physical Systems	2025	M.A. Ferrang	Generative AI in Cybersecurity: A comprehensive review of LLM Applications and Vulnerabilities.	GPT-4, BERT, RoBERTa, T5, LLaMA, Random Forest, LSTM, CNN, Transformer models, GANs, VAE, DBSCAN	• Comprehensive analysis of 35 LLM models showing GPT-4 achieved 96.25% accuracy on cybersecurity knowledge datasets. • Transformer-based models outperformed traditional approaches in malware classification, achieving state-of-the-art F1-scores of 0.6149 on benchmark datasets..	• Limited datasets lacking size and diversity for security tasks. • Need for more robust datasets tailored to cybersecurity applications and better domain-specific pre-training methodologies for enhanced LLM performance in security contexts.	• High computational requirements and vulnerability to adversarial attacks. • Models showed performance degradation on diverse datasets, and existing benchmarks may overestimate capabilities in real-world security-sensitive environments.	https://journal.esrgroups.org/jes/article/view/8329
IJARCCE (International Journal of Advanced Research in Computer and Communication Engineering)	2024	Likitha R, Tarun N, Pallavi N, Vidhey V Gaonkar	Enhancing Threat Detection: Integrating ELK-Based SIEM with IDS and Pattern Recognition Algorithms	ELK Stack (Elasticsearch, Logstash, Kibana), Suricata IDS, Pattern Recognition Algorithms, Machine Learning algorithms	• Integration achieved 96.5% detection rate with significant reduction in false alarms to 2.3% compared to 5% in conventional systems. • Mean response time of 1.2 seconds demonstrated near real-time threat detection and response capabilities.	• Limited evaluation against advanced persistent threats and lack of AI/ML integration for detecting unknown and emerging threats. • Insufficient testing on large-scale enterprise environments and diverse attack scenarios.	• Computational intensity and data quality issues affecting system performance. • Pattern recognition algorithm requires dimensionality reduction techniques and faces challenges with complex network environments and evolving attack patterns.	https://www.computer.org/csdl/magazine/co/2025/05/10970188/260SnWvsctW
IJRASET (International Journal for Research in Applied Science and Engineering Technology)	2024	Suraj Regi, Dr. Gurpreet Kaur	Improving Threat Detection with Elastic Search based Security Information and Event Management	Elastic Stack (Elasticsearch, Logstash, Kibana), Sysmon, Winlogbeat, VirusTotal integration, Arkime, T-Pot honeypot	• Elastic Cloud-based SIEM successfully provided centralized security operations with real-time monitoring and automated threat detection. • Integration with VirusTotal and Arkime enhanced overall threat detection capabilities through automated scanning and network packet analysis.	• Limited autonomous incident response capabilities and lack of AI engine integration. • Need for more comprehensive evaluation against sophisticated attack scenarios and scalability testing for larger enterprise environments.	• Resource allocation requirements for cloud-based deployment and dependency on external threat intelligence services. • Limited evaluation of system performance under high-volume attack scenarios and integration complexity with existing security infrastructure.	https://www.sciencedirect.com/science/article/pii/S2590005625000268

RESEARCH GAP

Current Limitations & Missing Solutions

- Lack of standardization in agent reasoning models, making it difficult to benchmark cognitive autonomy in cybersecurity AI agents.

- Limited empirical research on practical governance tools to prevent dual-use (misuse) of defensive agentic AI systems in cybersecurity.

- Early-stage exploration and limited adoption of quantum-safe federated learning for agentic systems, leading to insufficient resilience to quantum-enabled threats.

- Under-researched strategies for calibrating human trust in semi-autonomous defense agents, particularly in diverse operational and socio-technical contexts.

- Fragmented alignment between technical agentic AI architectures and regulatory compliance systems, resulting in difficulties with auditability and legal interpretation.

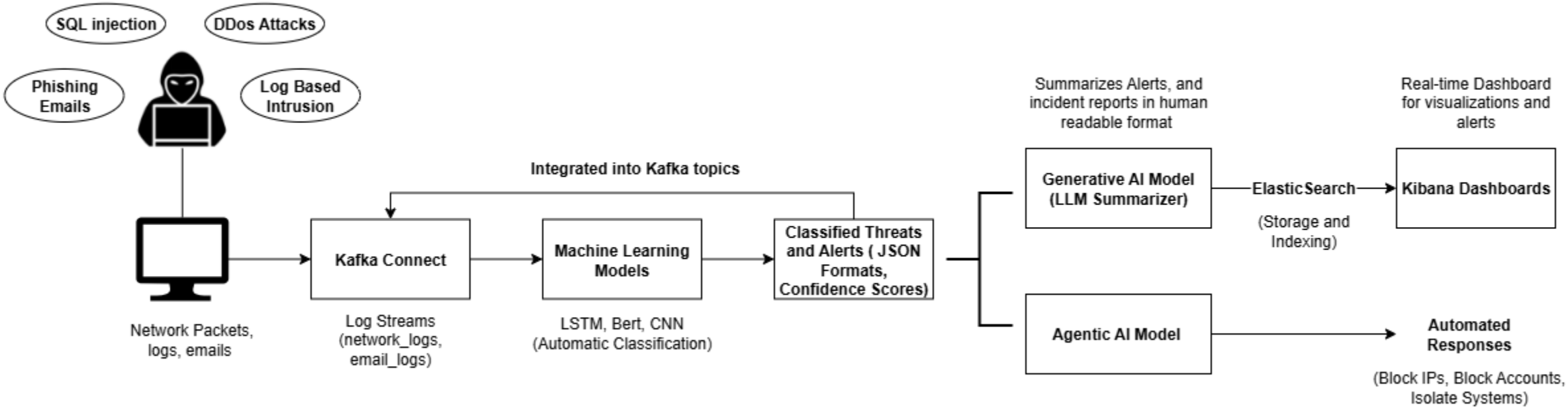
PROBLEM STATEMENT

Organizations struggle with real-time detection and response to sophisticated cyber threats. Existing tools can visualize alerts but lack autonomous threat interpretation and action. Our project aims to build an integrated pipeline that combines real-time data ingestion, AI-driven analysis, and automated response, improving detection accuracy and reducing response time.

OBJECTIVES

- **Real-time detection:** Continuous monitoring of cyber threats
- **Data ingestion:** Streaming pipeline using Kafka
- **AI-driven analysis:** Summarization and pattern recognition of alerts
- **Automated response:** Agentic AI for incident response
- **Accuracy improvement:** Reduce false positives in detection
- **Faster response:** Minimize incident handling time
- **Unified dashboard:** Visualize alerts, responses, and system health
- **Scalability:** adaptable design for enterprise-scale environments

PROPOSED METHODOLOGY



The system ingests network packets, logs, and emails through Kafka, where machine learning models (LSTM, BERT, CNN) classify potential threats. Classified alerts are summarized by a generative AI model, stored and indexed in Elasticsearch, and visualized in Kibana dashboards. An agentic AI model then executes automated responses such as blocking IPs, disabling accounts, or isolating compromised systems.

References

1 *N. Kshetri, Transforming Cybersecurity with Agentic AI to Combat Cyber Threats, unpublished manuscript, accessed Sep. 22, 2025. [Online].* Available: https://dl.acm.org/doi/10.1016/j.telpol.2025.102976

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3 *M.A. Ferrag*, *Generative AI in Cybersecurity: A Comprehensive Review of LLM Applications and Vulnerabilities, unpublished manuscript, accessed Sep. 22, 2025.* https://journal.esrgroups.org/jes/article/view/8329

4 *Likitha R, Enhancing Threat Detection: Integrating ELK-Based SIEM with IDS and Pattern Recognition Algorithms, unpublished manuscript, accessed Sep. 22, 2025.* https://www.computer.org/csdl/magazine/co/2025/05/10970188/260SnWvsctW

5 *Suraj Regi, Improving Threat Detection with Elastic Search-based Security Information and Event Management, unpublished manuscript, accessed Sep. 22, 2025.* https://www.sciencedirect.com/science/article/pii/S2590005625000268